

PART I: NUMBER SYSTEMS

NUMBER COMPLEMENTS

SAAD BAIG – DIGITAL LOGIC DESIGN



The animations and contents in these slides are originally adapted from the DLD course slides designed by Dr. Hasan Baig and Mr. Junaid Ahmed Memon. The original slides can be accessed from <https://www.hasanbaig.com/resources>

© Hasan Baig and Junaid Ahmed Memon. All rights reserved. These files and accompanying materials are made available under the terms of “Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0) License” and is available at <https://creativecommons.org/licenses/by-nc-nd/4.0/>



SUBTRACTION OPERATIONS

- Let's do some subtraction operations with binary numbers.
 - **Example 1:** $(110)_2 - (101)_2 = 001$ ■ In decimal $6 - 5 = 1$
 - **Example 2:** $(110)_2 - (110)_2 = 0$ ■ In decimal $6 - 6 = 0$
 - **Example 3:** $(110)_2 - (111)_2 = ?$ ■ In decimal $6 - 7 = -1$
- In the case of Example 3, we now have to deal with some representation of a negative number.



COMPLEMENTS OF NUMBERS

- Complements are used in digital computers to simplify the subtraction operation and for logical manipulation.
- There are two types for each base- r system.

Radix Complement

- r 's complement
- 2's complement for binary numbers
- 10's complement for decimal numbers

Diminished Radix Complement

- $(r - 1)$'s complement
- 1's complement for binary numbers
- 9's complement for decimal numbers



DIMINISHED RADIX COMPLEMENT

- For a number N in base- r having n digits, the $(r - 1)$'s, complement, or diminished radix complement, is defined as:

$$(r^n - 1) - N$$



9'S COMPLEMENT

- For decimal numbers, $r = 10$ and $r - 1 = 9$.
- 9's complement of N is $(10^n - 1) - N$
- Here, 10^n represents a number that consists of a single 1 followed by n 0's. E.g. $10^3 = 1000$.
- And $(10^n - 1)$ is a number represented by n 9's. E.g. $10^3 - 1 = 999$.
- It follows that the 9's complement of a decimal number is obtained by subtracting each digit from 9.
 - **Example 1:** The 9's complement of 546700 is $999999 - 546700 = 453299$
 - **Example 2:** The 9's complement of 012398 is $999999 - 012398 = 987601$



1'S COMPLEMENT

- For binary numbers, $r = 2$ and $r - 1 = 1$.
- 1's complement of N is $(2^n - 1) - N$
- Here, 2^n represents a number that consists of a single 1 followed by n 0's. E.g. $2^3 = (1000)_2$.
- And $(2^n - 1)$ is a number represented by n 1's. E.g. $2^3 - 1 = (111)_2$.
- The 1's complement of a binary number is obtained by subtracting each digit from 1.
 - **Example 1:** The 1's complement of $1011000 = 0100111$
 - **Example 2:** The 1's complement of $0101101 = 1010010$
- You will notice that you can obtain 1's complement just by flipping the binary digit.



RADIX COMPLEMENT

- For a number N in base- r having n digits, the r 's, complement, or radix complement, is defined as:

$$\begin{array}{ll} r^n - N & N \neq 0 \\ 0 & N = 0 \end{array}$$

- Comparing with the $(r - 1)$'s complement, we note that the r 's complement is obtained by adding 1 to the $(r - 1)$'s complement, since $r^n - N = [(r^n - 1) - N] + 1$.



10'S COMPLEMENT

- For decimal numbers, 9's complement of N is $(10^n - 1) - N$
- 10's complement would be $[(10^n - 1) - N] + 1$
- It follows that the 10's complement of a decimal number is obtained by taking the 9's complement of a number (subtracting each digit from 9) then adding 1 to it.
 - **Example 1:** The 10's complement of 546700 is $999999 - 546700 + 1 = 453300$
 - **Example 2:** The 10's complement of 012398 is $999999 - 012398 + 1 = 987602$
- **Alternate Method:** From left to right, leave all significant 0's. Subtract the first digit from 10 and all other higher significant digits from 9.



2'S COMPLEMENT

- For binary numbers, 1's complement of N is $(2^n - 1) - N$
- 2's complement would be $[(2^n - 1) - N] + 1$
- It follows that the 2's complement of a binary number is obtained by taking the 1's complement of a number (flipping each binary digit) then adding 1 to it.
 - **Example 1:** The 2's complement of 1011000 is $0100111 + 1 = 0101000$
 - **Example 2:** The 2's complement of 0101101 is $1010010 + 1 = 1010011$
- **Alternate Method:** From left to right, leave all significant 0's and the first 1 unchanged. Invert all other higher significant digits.

ACTIVITY



- The 2's complement of a binary number 0110111 is:
- The 10's complement of a decimal number 246700 is:
- Calculate the $(r - 1)$'s complement of a number $(2AB)_{16}$



SUBTRACTION WITH COMPLEMENTS

- Elementary school method → Borrow concept
 - Borrow 1 when minuend digit is smaller than subtrahend digit.
 - Efficient using paper-pencil but not on hardware.

$\begin{array}{r} 15 \\ 765 \\ - 39 \\ \hline 726 \end{array}$ Borrow and Pay Back Method Or Equal Additions	$\begin{array}{r} 515 \\ 765 \\ - 39 \\ \hline 726 \end{array}$ Borrow and Regroup Method Or Decomposition
---	---



SUBTRACTION WITH COMPLEMENTS

- Steps of subtracting two n-digit number $M - N$:
 1. Add the Minuend M to the r 's complement of the Subtrahend N : $M + (r^n - N)$
 2. If $M \geq N$, the sum will produce an end carry r^n , which can be discarded; what is left is the result $M - N$.
 3. If $M < N$, the sum does not produce an end carry and is equal to $r^n - (N - M)$, which happens to be the r 's complement of $(N - M)$.
 - To obtain the answer in a familiar form, take the r 's complement of the sum and place a negative sign in front.



SUBTRACTION WITH COMPLEMENTS

- **Example 1:** Using 10's complement, subtract $72532 - 3250$.

$M =$	72532	
$N =$	03250	→ To keep the number of digits same
10's complement of $N =$	+ 96750	
Sum =	169282	
Discard end carry $r^n (10^5) =$	- 100000	→ Step 2 as mentioned previously
Answer =	69282	



SUBTRACTION WITH COMPLEMENTS

- **Example 2:** Using 10's complement, subtract $3250 - 72532$.

$$M = 03250$$

$$N = 72532$$

$$\text{10's complement of } N = + 27468$$

$$\text{Sum} = 30718$$

$$(\text{10's complement of } 30718) = - 69282$$



To keep the number of digits same



Step 3 as mentioned previously



SUBTRACTION WITH COMPLEMENTS

- **Example 3:** Use 2's complement to subtract $X - Y$ and $Y - X$ where $X = 1010100$ and $Y = 1000011$.

$X - Y$

$$\begin{aligned} X &= 1010100 \\ \text{2's complement of } Y &= + 0111101 \\ \text{Sum} &= 10010001 \\ \text{Discard end carry } r^n (2^7) &= - 10000000 \\ \text{Answer } (X - Y) &= 0010001 \end{aligned}$$

$Y - X$

$$\begin{aligned} Y &= 1000011 \\ \text{2's complement of } X &= + 0101100 \\ \text{Sum} &= 1101111 \\ - (\text{2's complement of Sum}) &= - 0010001 \end{aligned}$$